

Online Utility Bill Payment System

Objective

The objective of this project is to develop a RESTful backend system for an Online Utility Bill Payment System using Spring Boot, MySQL, and Spring Data JPA.

The system allows:

- Users to register, view their bills, and make payments
- Admins to generate bills, and monitor users, bills, and payments

This service is backend-only, exposed via REST APIs, and testable using tools like Postman.

Technologies Used

Technology	Purpose
Java (JDK 8+)	Programming Language
Spring Boot	Backend framework
Spring Data JPA	ORM and repository abstraction
MySQL	Relational database
Maven	Build & dependency management
Postman	API testing

Modules / Functionalities

User Module

- Register a user
- View individual profile
- View user-specific bills
- View payment history

Bill Module

- Generate bill (Admin)
- View bills by:
 - User ID
 - Bill ID
 - Status (PENDING/PAID)
- Filter bills by user + status

Payment Module

- Pay for a bill
- Get payment details by bill
- Get all user payments
- Admin view of all payments

Admin Module

- Generate bills for users
- View all users
- View all bills
- View all payments

Entities and Relationships

User

- Fields: id, name, email, phone, address
- Role: USER or ADMIN
- Relationships: One-to-many with Bill

Bill

- Fields: id, amount, dueDate, status, category
- Status: PENDING or PAID
- Relationships: Many-to-one with User, One-to-one with Payment

Payment

- **Fields:** id, amountPaid, paymentDate, status
- **Status:** SUCCESS or FAILED
- **Relationship:** One-to-one with Bill

REST API Flow Summary

► User Endpoints (/api/users)

- POST /register → **Register new user**
- GET /get?id=1 → **Get user by ID**
- GET /all → **List all users**

► Bill Endpoints (/api/bills)

- POST /generate?userId=1 → **Generate bill for user**
- GET /byuser?userId=1 → **Get bills by user**
- GET /bystatus?status=PENDING → **Get bills by status**
- GET /byid?billId=1 → **Get bill by ID**
- GET /byuserstatus?userId=1&status=PAID → **Bills by user & status**

► Payment Endpoints (/api/payments)

- POST /pay?billId=1 → **Make payment**
- GET /byuser?userId=1 → **Payments by user**
- GET /bybill?billId=1 → **Payment for bill**
- GET /all → **Admin: all payments**

► Admin Endpoints (/api/admin)

- POST /generatebill?userId=1 → **Admin generates bill**
- GET /users → **View all users**
- GET /bills → **View all bills**
- GET /payments → **View all payments**

Project Execution Flow

1. User registers via /api/users/register
2. Admin generates a bill via
/api/admin/generatebill?userId=X
3. User views their bills via /api/bills/byuser?userId=X
4. User pays bill via /api/payments/pay?billId=Y
5. Admin monitors everything using:
 - /api/admin/users
 - /api/admin/bills
 - /api/admin/payments